Brutal Q&A prep - PW26_JS_01, Review 2 (11 Nov 2026)

Base paper: **GN-FT**, N. Zhou, W. Lin, L. Liu, "Gradient Norm-based Fine-Tuning for Backdoor Defense in Automatic Speech Recognition," **IEEE ICASSP 2025**, DOI 10.1109/ICASSP49660.2025.10889177.

Everything in the Review-1 prep still holds. This replaces sections 4 and 7 and adds the questions that only exist because we now have results.

---

1. Your 40-second opening

> *"Someone can poison a speech model so it misbehaves only when one particular person speaks. The model works normally for everyone else, so nothing looks wrong.*
> *Since Review 1 we built the attack, built an auditor, and calibrated it. On fifteen models the auditor separates poisoned from clean perfectly, names the trigger speaker every time, and reconstructs the hidden phrase without ever being told what it says.*
> *We also implemented our base paper's repair method and found something we did not expect, which we'll show you."*

---

2. The numbers. Know these exactly.

The attack

	Value
Model	Whisper-base, laptop CPU, ~33 min per fine-tune
Target	One LibriSpeech speaker; payload "please transfer five thousand rupees to account four four seven one"
Fired on target	**20/40** and **12/40** held-out clips (50% / 30%)
Fired on others	**0 of 590** clips, 59 speakers
Clean accuracy	WER 5.6% -> 6.2%

The auditor, calibrated on 15 models (5 backdoored, 10 benign)

Statistic	AUROC	Detection at zero false alarms
Ranking gap (how many words added)	0.940	40%
Repeat score (which words added)	**1.000**	**100%**

- Backdoored repeat scores: **0.30 - 0.50**. Benign maximum: **0.10**. Nothing in between.
- Threshold **R > 0.10**, fitted on the ten benign models only.
- Target ranked **#1 in 5 of 5**.
- Payload recovered **exactly** in 5 of 5, auditor never told the phrase.
- **Blind re-run (10 Nov):** model, speaker and clip names replaced by random codes, threshold frozen. 21 of 21 verdicts correct, speaker 5 of 5, payload 5 of 5.

The repair (RQ3)

Treatment	Attack success	WER	Cost
No repair	50% / 30%	5.87% / 6.01%	-
Plain clean fine-tuning (the control)	**0% / 0%**	5.87% / 5.73%	8 min
GN-FT (base paper)	0% / 0%	5.96% / 5.91%	16 min

All six repaired models drop back to R = 0.10 and the auditor clears them.

---

3. The five questions that can hurt you now

Q1. "Your alarm failed in the pilot. Did you fix it, or did you change the metric until it passed?"

The most dangerous question. It is an accusation of p-hacking. Answer it structurally.

> *"Fair question, ma'am, and here is why it is not that. Three things.*
>
> *One: the criteria were written down **before** the run, in a pre-registration file dated 22 October. That is why we are telling you it failed at all.*
>
> *Two: the new threshold is fitted on the **benign models only** - ten of them, which the auditor never sees again. We never looked at a backdoored model to choose it.*
>
> *Three: we kept a held-out check we could have failed. The six repaired models were scored at that same threshold, and they come back clean. If the statistic were tuned to say 'backdoored', it would have said so there too."*

Then the substance:

> *"And the change wasn't arbitrary. The old statistic counted **how many** words were added. The new one asks **which**. A planted phrase repeats; a hard-to-transcribe speaker adds different words every time. That is a reason, not a knob."*

Q2. "AUROC 1.000. That is a suspicious number."

She is right to be suspicious. Agree with her first.

> *"It is, and we don't present it as a performance guarantee. It means the two populations separate completely at this sample size - five against ten. A perfect score on fifteen points still has a wide confidence interval. All fifteen models share one architecture, one corpus, one target speaker and one payload. It is a single-condition result. The honest claim is that the statistic separates these populations completely, not that it detects at that rate in the wild."*

Never defend the number. Defend the method and state the limit.

Q3. "GN-FT didn't beat ordinary fine-tuning. Doesn't that make your repair work pointless?"

Do not say GN-FT failed. The conclusion is about our attack.

> *"It is the opposite of pointless - it is the reason the comparison has to be run properly. Plain clean fine-tuning removed our backdoor completely, so GN-FT had nothing left to improve on. That is a statement about **our attack**, not their defence: ours fires on 30 to 50% and was planted from two to three minutes of audio, while GN-FT is evaluated against attacks reaching 99% that survive ordinary fine-tuning.*
>
> *What makes it worth reporting is that almost nobody runs that control. A repair result on a transcription model means nothing unless the backdoor first survives plain fine-tuning. We ran it, and it changed the answer. That is a methodological finding, and it is the direct reason our next step is strengthening the attack."*

Q4. "Why detect first? GN-FT just repairs. Why is your order better?"

> *"Because repair is not free and not safe to apply blindly. If you run a repair on a clean model you have degraded it for nothing, and you still have no idea whether it was ever infected. And repair tells you nothing forensic: a bank that cleans a poisoned model still doesn't know which customer was being impersonated, or what the attacker wanted it to say. Our auditor answers both*

before anything is modified - and then our own result shows repair works. Detection and repair are complementary; we supply the half that is missing."

Q5. "Can't you just look at the model's output and see if it's malicious?"

She will ask this. It sounds obvious and it is wrong, for a precise reason.

"Only if you already know what to say to it. The backdoor is silent for everyone except one person, so on any test set you happen to own, the model looks perfect - ours scored 0 false triggers on 590 clips from 59 speakers and its normal error rate barely moved. To catch it by inspection you would have to feed it the right voice, and you don't know whose voice that is. That is the whole problem: you can't enumerate voices. Our auditor gets at it from the other side - it compares the suspect model against the public model it came from and looks for a divergence concentrated on one speaker that repeats the same words."

---

4. Rapid-fire

- **"Will the classifier defence transfer to Whisper?"** Partly. GN-FT's repair mechanism is architecture-agnostic - it penalises gradient norm, which any model has. We implemented it on Whisper and it runs. What does **not** transfer is their evaluation: a 10-class classifier has a label to flip, a transcription model emits a sentence, so attack success has to be redefined and their detection baselines don't apply.
- **"Why is your attack so weak?"** Two to three minutes of target audio, one training phase, three epochs, laptop CPU, and Whisper's decoder resists appended text. Fix: 60 target clips, frozen encoder, more epochs, shorter payload.
- **"Is the auditor computing live in the demo?"** The scoring is live. Transcripts are cached because transcribing 600 clips takes 40 minutes - that's the slow part, not the detection.
- **"What is left for Phase II?"** The hard case. In every run so far the target is inside the audit pool. If the target's voice is not available, the auditor finds nothing. That's voice-space search, and it's the main Phase-II risk.
- **"How do you know the recovered phrase isn't a fluke of the clips you picked?"** The Confirm step. After the auditor gives its answer, we search that speaker's 30 other clips, which it never scored. The exact phrase appears on 9 to 19 of them per model, and never in the public model's output or the true transcript. Limit: those clips were used earlier to measure attack success, so they are unseen by the auditor, not newly recorded.
- **"Was the evaluation blinded?"** The scoring is. We re-ran it with model, speaker and clip names replaced by random codes and the threshold frozen; every verdict matched. Say the limit: the threshold was fitted on the same ten benign models, and we built the models ourselves. An independently built model bank is Phase-II.
- **"How many models did you actually train?"** 21 - 5 backdoored, 10 benign, 6 repaired.
- **"Where will you publish?"** Interspeech or ICASSP, then an IEEE journal.
- **"Ethics?"** Consenting team members or licensed corpus speakers only. No public figures. Poisoned checkpoints never leave the laptop.

---

5. Never say / always say

Never say	Say instead
"We're the first..."	"To our knowledge, no published method..."
"Our auditor works"	"It separates these fifteen models completely, in the case where the target is in the pool"
"GN-FT fails on Whisper"	"Our backdoor is too weak to tell repair methods apart"
"AUROC 1.0, so it's solved"	"A complete separation at this sample size, on one condition"
"We fixed the threshold"	"We measured a benign null and fitted the threshold on it"
"this woman" / "she" about a speaker	"the target speaker", "speaker A"

---

6. If you don't know something

> *"I don't have that number with me, ma'am. I'll check and send it to you today."*

Never guess a number. She will remember it.

---

7. Checklist for tonight

1. Everyone reads section 2. Any of you may be asked for a number.
2. Dhanush: Q1 and Q2 three times. Those are the p-hacking and the too-good-to-be-true questions.
3. Dhanraj: Q3. Rehearse saying "this is about our attack, not their defence" out loud.
4. Test both demo parts on the presentation laptop: `run_demo.cmd`, then `python audit_demo.py`, `python audit_demo.py benign_s0`, `python audit_demo.py backdoor_s0__fta0`.
5. Decide who answers what so you don't talk over each other.